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AI-POWERED ATTENDANCE MANAGEMENT

Face Attendance System

Using Facial Recognition to Automate, Secure, and Modernize Attendance Tracking

Traditional Attendance Is Broken



Slow & Manual

Roll calls waste class and work time every single day.



Prone to Errors

Manual entry invites typos, omissions, and disputes.



Easy to Manipulate

Proxy sign-ins let people mark attendance for others.



Not Scalable

Paper and spreadsheets collapse under growing headcount.

Face recognition solves all four problems at once — with automation and accuracy.

What the System Delivers



Automates Attendance

No manual roll calls — check-in happens on its own.



Stops Proxy Attendance

Only real, verified faces get logged into the system.



Improves Accuracy

Consistent, error-free records every time.



Saves Time

Check-in completes in under a second, per person.



Gives Instant Insights

Real-time attendance data, ready whenever it's needed.

Face Recognition Is Already Everywhere



Smartphones



Airports



Security Systems



Online Banking



Attendance Systems

From unlocking a phone to boarding a flight — the same core technology now powers modern classroom and workplace attendance.

System Overview



Face Detection



Face Recognition



Attendance Logging



Dashboard Analytics

BUILT WITH

FastAPI

React

FaceNet ONNX

SQLite

The pipeline runs end-to-end: a captured face is detected, recognized against stored identities, logged with a timestamp, and instantly reflected on the dashboard.

Data Collection & Preparation

LFW DATASET

13,233

total images

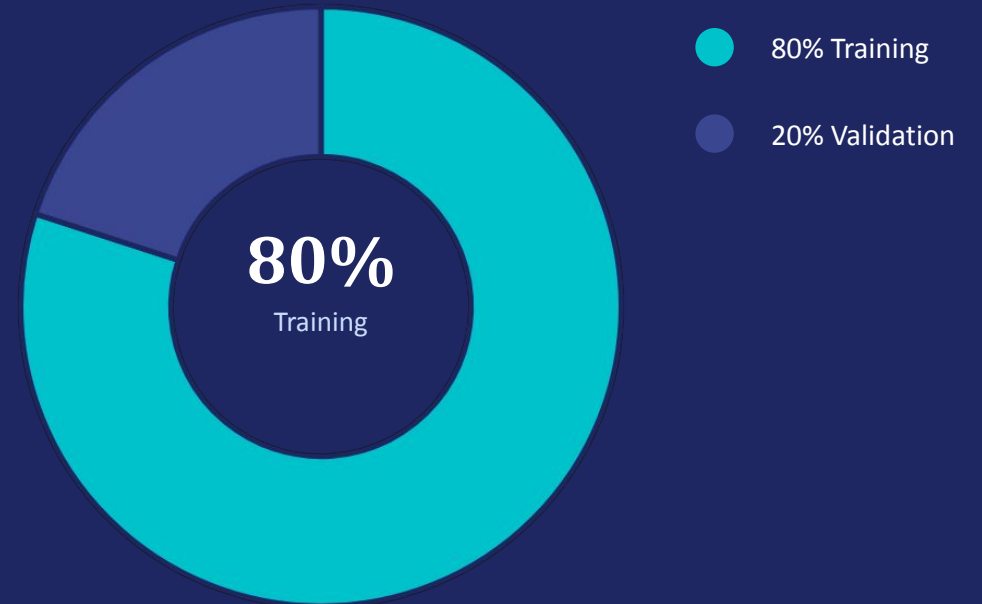
5,749

identities

Many identities had only 1–2 images, so the set was filtered to keep only identities with 5+ images — giving enough variation in:

- Angle
- Lighting
- Expression

TRAIN / VALIDATION SPLIT



Results & Metrics

90.39%

Accuracy

88.05%

Precision

89.63%

Recall

87.49%

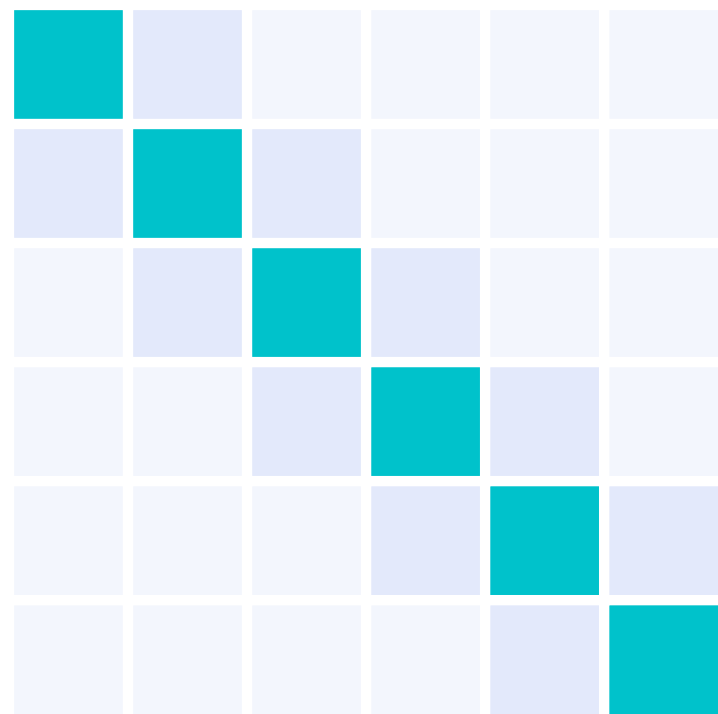
F1 Score



Strong identity separation

Evaluation metrics were computed and saved to `evaluation_report.txt` in the project root. The model achieves consistently high accuracy across identities, confirming it reliably tells people apart.

Confusion Matrix Insight



Illustrative diagonal-dominant matrix



Mostly Diagonal

The vast majority of predictions fall on the diagonal — strong correct classifications.



Very Few Errors

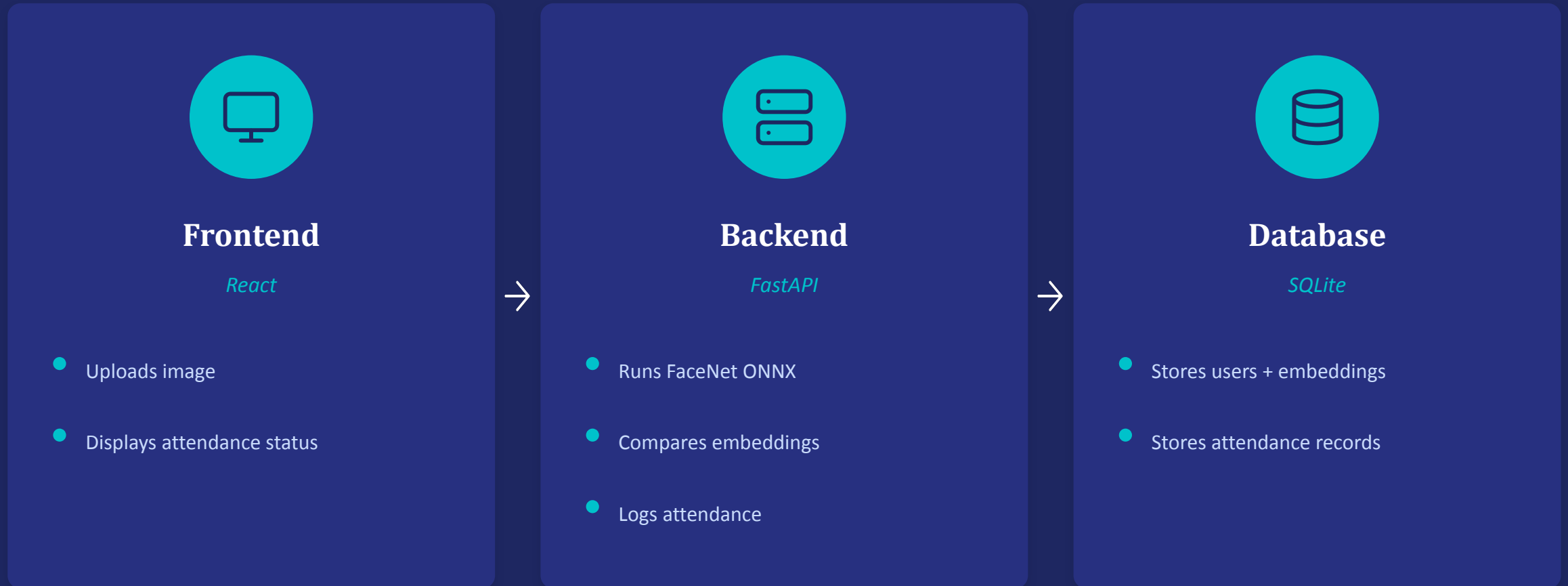
Off-diagonal values are minimal, meaning low misclassification across identities.



Stable Separation

Indicates the model reliably distinguishes identities across all classes.

System Architecture



Intelligent Components



Face Recognition Module

Automatically identifies students using facial embeddings and real-time detection.



AI Attendance Assistant

Lets users interact with attendance data through natural-language queries.

Together, these two components make the system intelligent, efficient, and user-friendly.

Face Detection & Alignment



Face Detection (MTCNN)

PURPOSE

- Detect face region
- Crop face
- Identify key landmarks

OUTPUTS

- Bounding box
- Facial landmarks



Face Alignment

Before recognition, the system adjusts each face for:

- **Angle**
- **Rotation**
- **Eye position**

This ensures consistent face inputs — which leads directly to higher recognition accuracy.

Feature Extraction — FaceNet ONNX



512-Dimensional Embedding

FaceNet converts each face into a mathematical fingerprint — a vector that uniquely represents that person's identity.

This embedding is what gets compared, stored, and matched throughout the entire system.

EXTRACTED FEATURES INCLUDE



Eye Distance



Jaw Shape



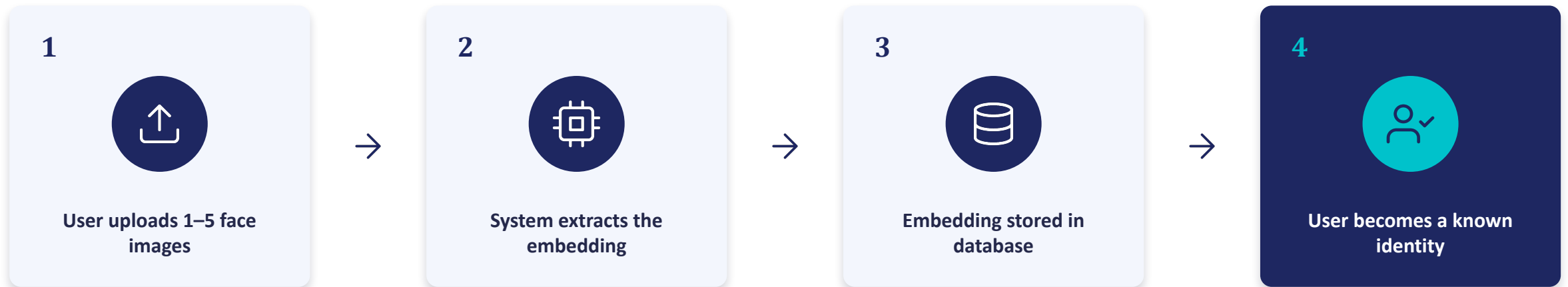
Nose Width



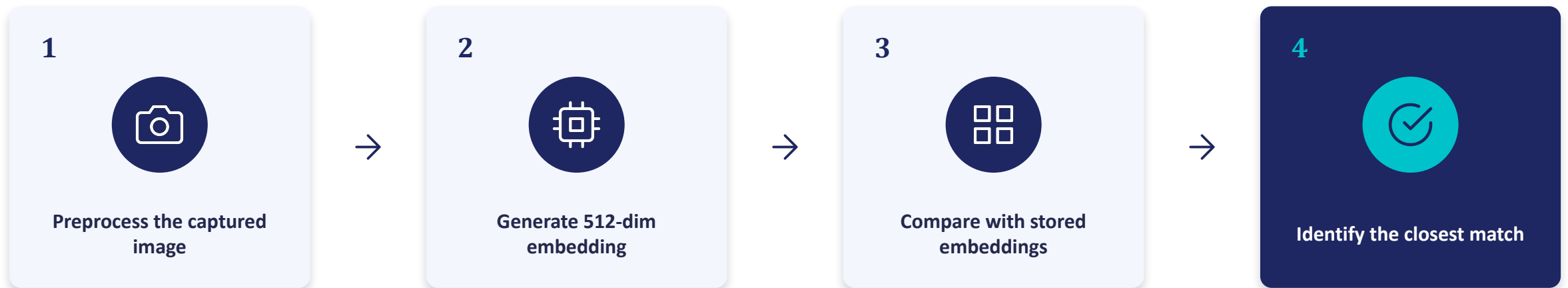
Skin Texture

POST /register/register

Registration Flow



Recognition Flow



Attendance Flow



Dashboard & Analytics



Total Attendance



Daily Attendance



Unique Users



Attendance Trends

API ENDPOINTS

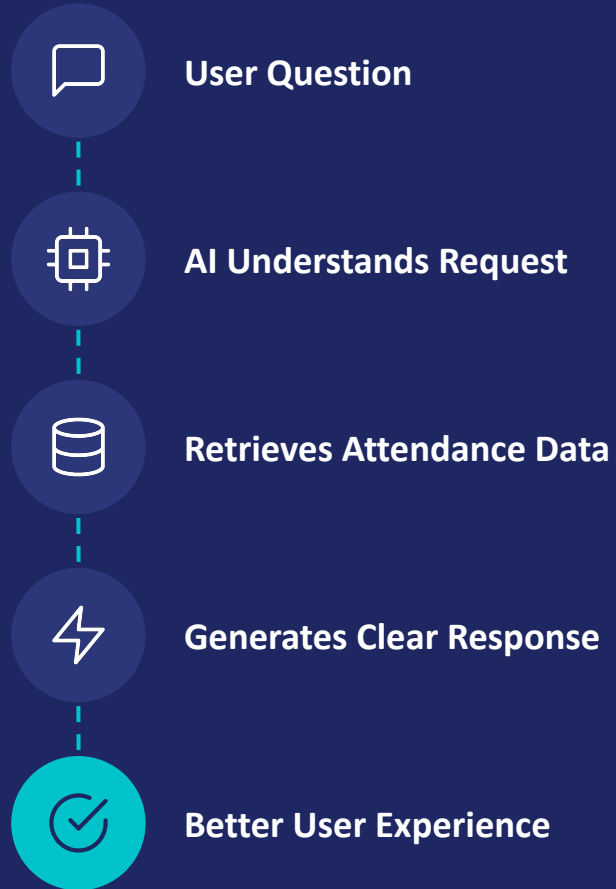
`/attendance/all`

`/attendance/analytics`

`/attendance/today`

Real-time visibility means administrators never wait for end-of-day reports to know what's happening.

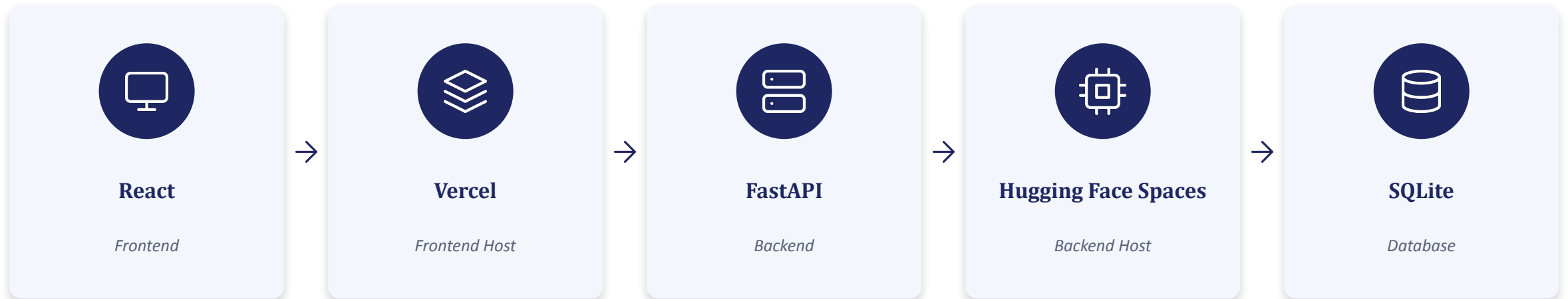
AI Attendance Assistant



KEY POINTS

- Enables users to interact using natural language.
- Retrieves attendance information from the backend.
- Provides quick answers without navigating multiple pages.
- Enhances the usability of the attendance system.

Technology Stack



Communication via REST APIs

All layers exchange data over standard REST endpoints, keeping the frontend, backend, and database cleanly decoupled and independently deployable.

Challenges & Solutions

CHALLENGE

OUR SOLUTION



Low-quality images

→ Face alignment + embedding normalization



Similar-looking students

→ High-dimensional FaceNet embeddings



Users need easy access

→ Natural-language AI Assistant

Key Learnings & Future Improvements



Multi-Camera Support

Expand recognition to multiple camera feeds for larger spaces and simultaneous check-ins.



Cloud Database (PostgreSQL)

Move from SQLite to a scalable cloud database for larger deployments and concurrent access.

Conclusion



Fast, automated attendance



High accuracy with FaceNet



Easy frontend integration



Scalable for classrooms, offices, events

The Face Attendance System delivers a faster, fairer, and smarter way to track presence.

SPECIAL THANKS



"Unless the Lord builds the house, the builders labor in vain."

— Psalm 127:1

Amighty God

Aperion AI and ML

For teaching us with dedication and guiding us throughout this journey.

Debela Yadeta

For guidance, mentorship, and support throughout this project.